2nd Island Health & Sustainability Conference

CONFERENCE PROCEEDINGS

Gump South Pacific Research Station, Moorea, French Polynesia
18–20 November 2025

Organized under the auspices of RESIPOL
French Polynesia's Research, Higher Education, and Innovation Consortium

Program Committee: Neil Davies (Chair), Hervé Bossin (ILM), Mayalen Zubia (UPF), Tamatoa Bambridge (CNRS), Guillaume Mitta (IFREMER), Adrienne Correa (UCB), Simon Ducatez (IRD)

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Executive Summary

The 2nd Island Health and Sustainability Conference convened from 18 to 20 November 2025 at the UC Berkeley Gump South Pacific Research Station in Moorea, French Polynesia. Organized under the auspices of RESIPOL, the French Polynesian Research, Higher Education, and Innovation Consortium, the conference brought together over 50 participants from local institutions, Pacific Island communities, and international research organizations to advance collaborative research addressing urgent sustainability challenges facing island social-ecological systems.

Building on the inaugural 2024 conference, this second edition focused on translating scientific knowledge into actionable strategies aligned with French Polynesia's Climate Plan and Innovation Strategy. The conference was structured around four interconnected missions: food systems transformation, renewable energy development through seaweed bioeconomy, land-sea resilience, and public health through elimination of vector-borne disease. These missions served as organizing frameworks for three days of keynotes, panel sessions, breakout discussions, and collaborative project development.

Day 1, themed "Island Bioeconomies: From Missions to Markets," opened with a keynote by Nils Gilman of the Berggruen Institute on planetary governance, followed by presentations from each RESIPOL member institution. Sessions explored the blue bioeconomy, including seaweed diversity, oyster aquaculture, biofuel life-cycle assessment, and blue carbon market standards. Breakout sessions allowed participants to develop mission-specific action plans.

Day 2, "Communities of Inquiry," examined the governance, educational, and social infrastructures needed to support island sustainability. A keynote by David Zilberman on bioeconomy and innovation was followed by sessions on biocultural stewardship, social license to operate, scholarly publication ethics, and educational programs including the Marine Educational Areas model and the proposed Ocean Environmental Leadership Program. The Social Capital Foundation led a workshop on impact measurement using theory-of-change frameworks.

Day 3, "Sites Ateliers," focused on place-based case studies ranging from the Tetiaroa Conservation and Sustainable Use Plan (CASUP) to the 4Site Pacific Transect Collaborative linking Tetiaroa, Moorea, Palmyra, and Coconut Island. Sessions highlighted Indigenous-led marine governance, the Tereheamanu community of communes, and a vision for mapping Indigenous seas across the Pacific.

Key themes that emerged across all three days included the critical importance of Indigenous and local knowledge in governance and conservation, the potential of seaweed as a transformative feedstock for the circular bioeconomy, the need for place-based data platforms that link research outputs back to communities, and the promise of islands as laboratories for innovative governance models that integrate planetary-scale challenges with local sovereignty. The conference produced concrete next steps in mission-specific project development, educational program design, and inter-institutional collaboration frameworks.

Day 1 (Tuesday 18 November) — Island Bioeconomies: From Missions to Markets

Opening and Contextualization

The conference opened with a traditional Polynesian welcome by Te Pu Atitia, grounding the proceedings in local culture and traditions. Neil Davies, Director of the Gump Station and conference chair, welcomed participants and introduced RESIPOL, the six-member consortium comprising the University of French Polynesia, IRD, CNRS, IFREMER, Institut Louis Malardé, and UC Berkeley. Davies emphasized the interconnection between local island identities and planetary systems, noting that the ocean and atmosphere connect local actions to global consequences.

Keynote: Island Earth — Governing in the Planetary Age

Nils Gilman — *Berggruen Institute*

Gilman presented the central thesis of his book, co-authored with Jonathan S. Blake, arguing that existing governance structures rooted in 17th-century concepts of state sovereignty are inadequate for addressing planetary-scale challenges such as climate change, biodiversity loss, and ocean health. He introduced the concept of "planetary sapience" — the growing capacity of sensor networks, data systems, and algorithms to understand planetary phenomena in ways never previously possible.

He proposed "functional sovereign entities" that would operate across national boundaries on specific issues such as carbon budgets, citing the IAEA's nuclear inspection regime as a precedent for segmented sovereignty. Gilman advocated for multi-species flourishing as the ethical ideal for governance, discussing how technologies such as cetacean communication monitoring could give political voice to non-human others. He concluded by highlighting small islands like Moorea as particularly fruitful laboratories for modeling these new governance approaches, given their defined boundaries and existing integration of local and global knowledge systems.

Session: FP Health and Sustainability Research (RESIPOL)

Chaired by Neil Davies, this session featured presentations from each RESIPOL founding member institution, providing an overview of current research priorities in French Polynesia.

Innovative Approaches to Vector-Borne Disease

Hervé Bossin — *Institut Louis Malardé*

Bossin presented ILM's work on innovative approaches to controlling mosquito-borne diseases in French Polynesia, where dengue, Zika, and chikungunya remain persistent public health

threats. The presentation outlined the Sterile Insect Technique (SIT) and related genetic approaches being developed to suppress Aedes aegypti populations.

Exploring Tropical Seaweed Diversity for a Sustainable Bioeconomy

Mayalen Zubia — *University of French Polynesia (SECOPOL)*

Zubia presented a comprehensive overview of tropical seaweed diversity in French Polynesia, which harbors approximately 531 species (320 red, 119 green, and 92 brown). She emphasized that tropical seaweeds are among the least-studied marine organisms in coral reef ecosystems, with many regions remaining unexplored and high levels of endemicity (exceeding 60% for some genera in Pacific islands).

The presentation highlighted the cultural significance of seaweeds in Pacific Island traditions, including their use in purification rituals, traditional medicine, and mythology. Zubia described the multi-marker DNA barcoding approach being applied across French overseas territories to improve tropical seaweed reference databases. She also presented research on bioactive compounds from seaweeds, including antioxidant, antimicrobial, and anti-inflammatory properties, with applications in nutraceuticals, cosmetics, and biostimulants for agriculture.

Te Ora Fenua | Te Mauri Ora

Adrienne Correa — *UC Berkeley*

Correa presented her program on coral reef health and marine biodiversity research in French Polynesia, emphasizing the integration of genomics and environmental monitoring. She discussed the importance of working with local communities and the ethical considerations of genomic data collection, noting ongoing collaboration with Jane Anderson on data governance frameworks.

Advancing Tropical Oyster Farming: Sustainable Nutrition and Food Security

Cristian Monaco and Guillaume Mitta — *IFREMER Tahiti*

Monaco presented IFREMER's work on developing sustainable oyster aquaculture in French Polynesia as part of a broader strategy for food system transformation. He highlighted that 80% of food products are imported into French Polynesia, with over 50% of total emissions attributed to transport — compared to less than 25% globally.

The presentation detailed the identification of two native Polynesian species (Saccostrea echinata and S. cucullata) through genetic analysis and the optimization of hatchery production techniques. Successful grow-out trials achieved commercial size after 16 months for S. echinata. A solar-tide system was developed to combat Polydora parasite infestation, and research demonstrated the species' high capacity to cope with climate change.

Zone Atelier FP: Long-Term Social-Ecological Research

Tamatoa Bambridge — *CNRS — CRIOBE, Rahui Center*

Bambridge presented the Zone Atelier French Polynesia program, a long-term social-ecological research initiative under the CNRS framework. The program focuses on understanding the interplay between social systems and ecological dynamics across French Polynesian archipelagos, with emphasis on traditional governance systems (rahui) and contemporary marine management.

Towards a Strategy to Reduce Biodiversity Loss and Restore Biodiversity in French Polynesia

Simon Ducatez — *IRD Tahiti (SECOPOL)*

Ducatez presented the urgent case for a comprehensive biodiversity strategy in French Polynesia, noting that the territory is a hotspot of the global extinction crisis: 8.5% (14 species) of all bird species that went extinct globally over the past 500 years were French Polynesian endemics. Today, 15 of 26 endemic bird species are endangered or critically endangered.

He analyzed the failures of global biodiversity targets (Aichi Targets) and argued for locally adapted, measurable objectives for French Polynesia. Key questions include prioritizing which ecosystems and species to restore, defining reference models for restoration, involving communities in co-constructing projects, and avoiding “green colonialism.” The presentation outlined a prioritization framework balancing impact, importance, and feasibility.

Session: Introduction to IH&S Missions

Chaired by Thierry Lison de Loma, this session introduced the four strategic missions that frame the conference’s collaborative agenda.

Eliminating Mosquito-Borne Disease

Craig Montell — *UC Santa Barbara*

Montell presented cutting-edge genetic research aimed at improving the Sterile Insect Technique for mosquito control, with direct application to French Polynesia. His laboratory has identified the trpVa gene as critical for hearing and mating in Aedes aegypti mosquitoes — males with mutated trpVa are completely deaf and unable to fertilize wild-type females.

Building on this discovery, the team demonstrated that impairing vision (through trp gene mutation) paradoxically increases male mating competitiveness, likely by reducing distraction. This finding opens a path to engineering sterile males with enhanced mating drive to overcome a key limitation of conventional SIT programs. Montell also presented work on a safe, long-lasting insect repellent compound, offering a complementary approach to population suppression.

Land-Sea Resilience

Tamatoa Bambridge and Simon Ducatez — *CRIOBE / IRD*

Bambridge and Ducatez jointly presented the Land-Sea Resilience mission, focusing on the continuum between terrestrial and marine environments that is central to French Polynesia's 124 islands with approximately 4,500 km of coastline. Economic activities including tourism, fishing, aquaculture, and agriculture depend directly on the health of this continuum.

Ducatez reported on a workshop held at UPF in September–October 2025 with 50 participants that identified key monitoring parameters, inventoried 39 long-term monitoring programs and 11 databases, and produced GIS maps of monitored sites and watersheds on Tahiti and Moorea. Future directions include the COCOPOL project, a multi-million-euro research program on contaminants in the land-sea continuum.

Food Systems: Transforming Island Food Systems by 2030

Thierry Lison de Loma — *Vaihuti Fresh*

Lison de Loma presented the Food Systems mission with a 5–10-year horizon aligned with French Polynesia's strategic development agenda. The mission addresses heavy reliance on imported food, climate instability, new pests and diseases, and fragmented value chains. Target sectors include schools and tourism, which offer high leverage and accessible funding. Ethical foundations include local production and short circuits, organic agriculture (BioPasifika certification), responsible fisheries, cultural landscape conservation, and public health nutrition.

Energy: The Pathway to Decarbonize FP Using Seaweed

Thiago Correa — *UC Berkeley*

Correa mapped the potential for seaweed-based biofuels to displace French Polynesia's approximately 350 million liters of annual fossil fuel imports. He outlined how macroalgae cultivation could produce ethanol and other biofuels with significantly lower carbon intensity than conventional fossil fuels, while avoiding the land-use conflicts associated with terrestrial biofuel feedstocks.

Keynote: International Bioeconomy and Macroalgae Center

Indonesia and Tropical Seaweed: Challenges and Implications for French Polynesia

John Coates — *UC Berkeley*

Coates presented the vision for an international bioeconomy center focused on macroalgae, drawing lessons from Indonesia as the world's leading tropical seaweed producer. He discussed the challenges and opportunities of scaling seaweed cultivation and processing, and potential implications for developing a seaweed industry in French Polynesia. His presentation connected

the upstream production challenges to the downstream biorefinery concepts, including the potential for seaweed-derived biofuels, bioplastics, and food products.

Keynote: Blue Natural Capital

Blue Natural Capital and Whale Conservation

Mere Takoko — *Pacific Whale Fund*

Takoko introduced the concept of blue natural capital through the lens of whale conservation, discussing the ecological and economic value of cetacean populations in the Pacific. She highlighted whales' role in nutrient cycling and carbon sequestration, and advocated for governance frameworks that recognize the rights and personhood of marine species.

Session: Te Ara Moana — Blue Pathways to Sustainability

Chaired by Larry Tchiou, this session explored diverse approaches to building a blue bioeconomy.

Life-Cycle Assessment: Biofuel from Macroalgae in French Polynesia

David Scolari — *UC Berkeley (PhD Student)*

Scolari presented a life-cycle assessment framework for evaluating seaweed-based biofuel production in French Polynesia. He explained LCA methodology as an accounting of environmental impacts from "cradle to grave," noting that conventional fossil fuels have a carbon intensity of approximately 92–97 gCO2e/MJ. Marine-feedstock biofuels from seaweed avoid the land-use competition, freshwater requirements, and fertilizer inputs associated with terrestrial biofuel crops, potentially offering a low-carbon alternative for French Polynesia's energy sector.

Blue Bioeconomy in Aotearoa New Zealand: Lessons and Opportunities

Nigel Bradly — *Envirostrat New Zealand*

Bradly presented New Zealand's experience in catalyzing blue economy innovation, describing a model that progresses from origination through venture development, catalytic investment, proof of concept, and commercialization. Specific ventures include Save Our Scallops (combining restoration, aquaculture, and wild fishery management), ORA Reefs (barren kina removal, species reseeding, and artificial reefs), and Blue Nature Aotearoa (ocean leadership mentorship and student-led impact projects). The presentation offered frameworks for collaboration with French Polynesia.

Land-Sea Food Systems: Taro and Carbon Sequestration

Jenny Kahn — *William & Mary College*

Kahn, an archaeologist, presented research on traditional taro cultivation systems and their potential role in carbon sequestration. She examined the historical relationship between taro agriculture and marine ecosystems, demonstrating how traditional Polynesian land management practices created productive and resilient food systems with co-benefits for climate mitigation.

Marine Resources and Island Innovation Strategies

Cedrik Lo — *Fisheries Agency, DRMM (French Polynesia)*

Lo presented French Polynesia's fisheries management perspective and innovation strategies for marine resource use, connecting government policy frameworks to the conference's bioeconomy themes.

Breakout Sessions: IH&S Missions

Facilitated by Erin Robinson, participants divided into four mission-oriented groups for collaborative project development. Each group included a core team of domain specialists joined by rotating experts in economics, governance, education, and technology. The breakout sessions covered: (1) Regenerative Food Systems including aquaculture and agriculture, led by Thierry Lison de Loma and Cristian Monaco; (2) Seaweed Biofuel and Energy, led by Thiago Correa and Larry Tchiou; (3) Land-Sea Restoration and Resilience, led by Simon Ducatez and Jean Wencélius; and (4) Eliminating Vector-Borne Infectious Disease, led by Hervé Bossin.

Keynote: Blue Carbon Market Standards

Blue Carbon Market Standards

Adrien Conte — *IRD, La Réunion*

Conte presented the current state of blue carbon market standards, examining the frameworks for valuing and trading carbon credits from marine and coastal ecosystems. This keynote connected the conference's bioeconomy and resilience themes to potential financing mechanisms for island sustainability initiatives.

Day 2 (Wednesday 19 November) — Communities of Inquiry

Keynote: Bioeconomy, Innovation, and Environmental Leadership

David Zilberman — *UC Berkeley*

Zilberman provided a comprehensive framework for understanding the bioeconomy, distinguishing three core elements: the use of natural resources to produce food, fuel, and chemicals; the application of modern technologies (biotechnology, nanotechnology, information technology); and sustainable development that protects future generations' welfare while adapting and co-evolving with nature.

He categorized bioeconomy types including biotechnology-driven, resource-oriented, blue bioeconomy (using ocean resources), circular bioeconomy (waste valorization), and eco/green bioeconomy. Zilberman emphasized that the bioeconomy is built supply chain by supply chain, distinguishing between innovation supply chains (discovery, development, and upscaling) and product supply chains (upstream sourcing, midstream conversion, and downstream distribution). Constraints span technological, economic, policy, environmental, social, and geopolitical dimensions.

Session: Governance and Stewardship

Chaired by Tamatoa Bambridge, this session explored diverse governance models relevant to island sustainability.

Biocultural Restoration, Stewardship, and Governance

Kawika Winter — *Hawaii Institute of Marine Biology / Limahuli Garden*

Winter presented the concept of biocultural restoration as an extension beyond ecological restoration, focusing on social-ecological systems rather than ecosystems alone. Drawing on the He'eia National Estuarine Research Reserve in Hawaii, he demonstrated how Indigenous resource management practices constitute "ecomimicry" — sophisticated techniques including habitat engineering, trophic management, and fecundity management that produced sustainable resource abundance (āina momona).

Historical evidence shows that Indigenous Hawaiian aquaculture systems (loko i'a) increased reef fish abundance. Winter advocated for biocultural restoration that integrates structure, function, and diversity across the mountain-to-sea gradient, using traditional ecological knowledge as both scientific insight and governance foundation.

Social License to Operate: Challenges for the Seaweed Industry

Alexandra (Ali) Hill — *UC Berkeley*

Hill presented research on Social License to Operate (SLO) in the context of California's emerging seaweed farming sector. SLO is an informal, non-legal concept referring to ongoing community acceptance or approval. Using a pyramid framework from rejection to endorsement, the research identified three main barriers in the seaweed sector: regulatory and permitting challenges, environmental uncertainty, and community acceptance and awareness. The presentation provided actionable strategies for building legitimacy and trust through stakeholder engagement.

Rights and Representation, Including More Than Human

Jane Anderson — *NYU / Local Contexts*

Anderson addressed the legal and ethical frameworks for recognizing rights of nature and non-human entities. She discussed the Local Contexts initiative and its system of Traditional Knowledge and Biocultural Labels, designed to support Indigenous data sovereignty and research governance.

Whales: From Personhood to Indigenous Governance

Mere Takoko — *Pacific Whale Fund*

Takoko expanded on her Day 1 presentation, discussing the movement toward legal personhood for whales and how Indigenous governance frameworks in the Pacific provide models for non-human rights recognition. She presented the Moananui initiative for Indigenous-led marine sanctuaries.

Session: Informing Decision-Makers — Ethics and Technology in Scholarly Publication

Chaired by Neil Davies, this session addressed the infrastructure needed for trustworthy, place-based research data ecosystems.

Mapping Research Outputs: The Tetiaroa Use Case

Benoît Stoll and Tobias Fischer — *University of French Polynesia (SECOPOL)*

Stoll and Fischer presented their geospatial research on Tetiaroa atoll, including new equipment acquisitions (dGPS, multispectral drone, tide gauge) and a time series of satellite imagery from 1955 to 2025. Their analysis demonstrated that 10 of 13 motu on Tetiaroa remain stable, with no global warming effect detected on shorelines to date. Fischer presented an innovative machine-learning pipeline using XGBoost classification for automated shoreline extraction, with transfer learning capabilities that allow models trained on one atoll to be applied to others without retraining.

Documenting Research Compliance

John Marchioni — *Senior Advisor, iPlaces Alliance*

Marchioni presented approaches to documenting research compliance relevant to the UN BBNJ (Biodiversity Beyond National Jurisdictions) treaty. He proposed that the Standard Batch Identifier system for marine genetic resources should leverage existing local systems rather than imposing top-down identity management, using sealed envelopes of metadata that preserve local control over data and user management while providing integrity and trust for relying parties.

Accelerating the Translation of Research to Education

Warren Essey — *Kudu (remote)*

Essey presented Kudu's AI-powered educational platform, which replaces traditional textbooks with an integrated digital system anchored to high-quality content. The platform uses AI to help instructors create custom course materials through an iterative human-AI co-creation process. UCLA's first AI-curated course demonstrated improved student engagement. Future developments include KAI, a course-specific AI assistant that can provide instant feedback, generate insights for instructors, and create individualized content.

iPlace(s) Data Trust: A Place-Based Research Platform

Dave Vieglais — *NCEAS / DataONE*

Vieglais presented the iPlaces Alliance, a platform designed to address the disconnect between place-based research outputs and local communities. The platform tracks research applications, facilitates dialogue with local stakeholders, links research outputs back to place, and supports Indigenous data governance through Local Context Labels. It addresses "contextual collapse" — the problem where research outputs lose connection to the scientific, ethical, legal, and social contexts crucial for interpretation and fair use.

Keynote: Atitia Center — 20 Years of Culture and Education

Twenty Years of Culture and Education

Hinano Murphy — *Te Pu Atitia*

Murphy presented the two-decade journey of the Atitia Center in Moorea, highlighting its role in preserving and transmitting Polynesian cultural knowledge through educational programs that integrate traditional ecological knowledge with contemporary science.

Session: Education and Training — Curriculum Research and Development

Chaired by Maheata White Davies, this session explored diverse educational models for building sustainability capacity.

Educational Marine Areas (AME) in French Polynesia

Vainui Marakai — *DGEE (French Polynesia)*

Marakai presented the Educational Marine Areas program, an innovative model born in French Polynesia in 2012 that has become an international reference in environmental education. The network now includes 29 EMA zones managed by 31 schools with over 2,500 students across four archipelagos. The program is built on three foundations: knowing the coast and sea, living the coast and sea, and transmitting knowledge. Since 2016, the model has expanded to mainland France, reaching over 200,000 students in more than 1,500 educational areas.

Workforce Development in the Bioeconomy: Lessons from California

Alexandra (Ali) Hill — *UC Berkeley*

Hill drew lessons from California's agricultural sector for building a seaweed bioeconomy workforce. She used the baby carrot innovation story to illustrate how new product forms can transform entire industries, and discussed the value of circularity where farm waste becomes revenue streams. The presentation emphasized that stakeholder engagement, workforce development, and social sustainability are essential to moving from research to impact.

Ocean Environmental Leadership Program

Larry Tchiou, Maheata White Davies, and Thierry Lison de Loma — *UC Berkeley / Gump Station*

The team presented the proposed Ocean Environmental Leadership Program (Ocean ELP), a strategic extension of UC Berkeley's 25-year Beahrs Environmental Leadership Program. Ocean ELP would be anchored in Moorea and focused on marine and coastal systems, building capacity among mid-career island and ocean leaders from the Pacific Islands region. The program addresses a gap in existing environmental leadership training, which rarely focuses specifically on ocean systems with a Pacific, tropical-island grounding.

Panel: Impact Framework

Impact Measurement and Management

Laurence (Lolo) Kleykamp — *Social Capital Foundation*

Kleykamp led a workshop on impact measurement and management from a funder's perspective. She introduced theory-of-change frameworks as roadmaps for how and why initiatives will create desired outcomes, distinguishing between inputs, activities, outputs, outcomes, and impact. Participants worked through theory-of-change exercises for the

conference's four missions, beginning with the mosquito elimination project as a worked example.

Unconference: Project Development

The afternoon featured self-organized unconference sessions for collaborative project development. Topics included Blue Carbon (led by Tamatoa Bambridge), the Mozeroa mosquito project (led by Hervé Bossin), the iPlaces platform (led by Erin Robinson), and additional emerging projects. These sessions provided dedicated time for cross-disciplinary teams to develop specific project proposals, identify funding opportunities, and establish collaboration agreements.

Day 3 (Thursday 20 November) — Sites Ateliers

Keynote: Tetiaroa Conservation and Sustainable Use Plan — The First 10 Years

Stanley Rowland and Frank Murphy — *Tetiaroa Society*

Rowland and Murphy presented the first comprehensive review of the Tetiaroa Conservation and Sustainable Use Plan (CASUP), which was established to realize Marlon Brando's vision of creating a model of sustainable tourism and conservation. CASUP is unique in being voluntary, delegated to a nonprofit (Tetiaroa Society), and reliant on expert guidance with a united stakeholder vision.

The review included results from both French-language (11 responses from local stakeholders) and English-language (29 responses from technical experts) surveys. Local perspectives emphasized community engagement, stakeholder coordination, and cultural integration, while international experts focused on funding, enforcement, measurable outcomes, and research capacity. The presentation set the stage for CASUP 2.0 planning.

Session: Place-Based Policy and Management — 4Site Network

Chaired by Jayna DeVore, this session explored the network of place-based research sites across the Pacific.

Tetiaroa CASUP 2.0: The Next 10 Years

Tj Tate — *Tetiaroa Society*

Tate presented the planning process for CASUP 2.0, including the establishment of a steering committee, a comprehensive review of the original plan, survey results, and a timeline for development. The presentation outlined areas of consideration for the next decade, building on lessons learned from the first ten years of conservation management.

Moorea and Tahiti: Tereheamanu Community of Communes

Jean Wencélius and Tamatoa Bambridge — *CNRS — CRIOBE*

Wencélius and Bambridge presented the Tereheamanu community of communes in southern Tahiti, which became a pilot site in 2025. Following citizen consultation workshops in April 2025, the community identified essential needs spanning drinking water access, addiction prevention, energy independence, health, and food sovereignty. The project envisions a territory of resilience, integrating socio-ecological perspectives through action-research and co-management approaches.

Hawaii: Research Station Networks

Megan Donahue and Michael Bruno — *Hawaii Institute of Marine Biology / University of Hawaii*

Donahue and Bruno presented the Hawaiian component of the 4Site network, centered on Coconut Island (HIMB). They discussed the challenges and opportunities of coordinating place-based research across multiple institutions and jurisdictions.

4Site: From Transects to Knowledge Transects

Alexander Mawyer — *University of Hawaii (remote)*

Mawyer proposed extending the concept of ecological transects to "knowledge transects" that sample how observations, models, and practices are distributed across islands, disciplines, and communities. He argued that 4Site transects should serve as convergence infrastructure linking Indigenous ecological knowledge, social science, oceanography, genomics, and modeling. Knowledge transects would be explicitly FAIR and CARE-compliant, anchoring each data point in place, history, and power relations while enabling high-throughput analytics.

Keynote: Honu Submarine International Sea Station Tahiti

Honu Submarine International Sea Station

Richard Bailey — *Tetiaroa Society / Pacific Beachcomber*

Bailey presented the vision for the Honu Submarine International Sea Station Tahiti, an innovative underwater research platform that would advance marine science capabilities in French Polynesia.

Session: Blue Bioeconomies at Ocean Scales

Chaired by Tamatoa Bambridge, this session expanded the geographic and conceptual scope to encompass Indigenous-led marine governance across the Pacific.

Voice of the Ocean

Hinano Murphy — *Te Pu Atitia*

Murphy presented on ocean stewardship from an Indigenous Polynesian perspective, emphasizing the cultural and spiritual connections between Pacific peoples and the sea.

Traditional Council of Rapa Nui

Hetereki Huke — *Conseil Coutumier Rapa Nui*

Huke presented the traditional governance systems of Rapa Nui (Easter Island), providing a powerful example of Indigenous authority over marine and terrestrial resources. The presentation traced governance from the historical period through colonial administration to contemporary efforts to reclaim traditional decision-making authority.

Pacific Islands Development Program

Mary Therese Hattori — *East-West Center*

Hattori presented the Pacific Islands Development Program (PIDP), which since 1980 has assisted Pacific Islands leaders through capacity building, leadership interchange, and policy-relevant research. As a member of the Council of Regional Organisations of the Pacific (CROP) and secretariat for the Pacific Islands Conference of Leaders, PIDP provides a regional platform for advancing sustainability, gender equality, youth engagement, and culturally sustaining professional practices.

Mapping Indigenous Seas

Tamatoa Bambridge, Alexander Mawyer, and Jean Wencélius — *CNRS-CRIOBE / University of Hawaii*

Bambridge and Mawyer presented research revealing critical gaps in how Indigenous and local marine areas are represented in global conservation databases. Comparing the World Database on Protected Areas (WDPA) data for French Polynesia (6 points, 114 km^2, all from 1971–1992) with their own data (252 active features totaling 27,413 km^2), they demonstrated massive underrepresentation of actual marine management.

The presentation identified three fundamental problems: jurisdictional overlap between Indigenous, national, and international management regimes; land-sea overlap where marine protected areas cover terrestrial areas; and “unknown unknowns” representing the intentional or accidental spatial invisibility of management practices. They argued that bioeconomic models built on fixed boundaries and extractive value logics cannot accommodate fluid territories, relational governance, or multispecies obligations, calling for a fundamental epistemic shift toward plural, bioculturally-grounded bioeconomies.

Concurrent Sessions

The afternoon featured concurrent sessions including the Tetiaroa CASUP-2 Committee Meeting (co-chaired by Tj Tate and Stanley Rowland) and an iPlaces Planning session (led by Erin Robinson), which focused on developing the technical and governance architecture for the place-based research data platform.

Conclusions and Next Steps

The conference concluded with a plenary session identifying key outcomes and action items. Major themes for follow-up included advancing the four mission areas through specific project proposals and funding applications, developing the Ocean Environmental Leadership Program for launch from Moorea, continuing the CASUP 2.0 planning process for Tetiaroa, expanding

the 4Site network and knowledge transect concept, building the iPlaces platform infrastructure, deepening partnerships between RESIPOL institutions and international collaborators, and mapping Indigenous marine governance across the Pacific.

Appendix: Conference Participants

The following individuals participated in the 2nd Island Health and Sustainability Conference:

Adrienne Correa (UC Berkeley)
Alexander Mawyer (University of Hawaii)
Alexandra E. Hill (UC Berkeley)
Alida Kropf (ILM)
Benoit Stoll (University of French Polynesia)
Cedrik Lo (Fisheries Agency, DRMM PF)
Craig Montell (UC Santa Barbara)
Cristian Monaco (IFREMER Tahiti)
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